

Algebraic Geometry and Schubert Calculus

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April 2025

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1 Introduction

The objective of this report is to lay out the fundamentals of Schubert calculus.

The namesake of Schubert Calculus, 19th century German mathematician Hermann Schubert, solved various problems relating to the intersections of geometric objects.

At the 1900 International Congress of Mathematicians in Paris, German mathematician David Hilbert presented a list of unsolved problems that he considered important to the development of mathematics. The list had a large influence on the development of 20th century mathematics. In particular, the 13th problem on this list was to give a “rigorous foundation of Schubert’s enumerative calculus”. The search for such a rigorous foundation, among other motivations, led to the development of the fields of algebraic topology and algebraic geometry.

This report aims to lay out the modern understanding of Schubert calculus, and as such will require results from classical algebraic geometry and algebraic topology. In particular we will borrow heavily from [7] for algebraic geometry and [5] for algebraic topology. We will also refer to [2] for some results in the study of symmetric groups and [6] and [8] for Schubert calculus itself.

2 Preliminaries

Let us first lay out some of the results from algebraic geometry that will be necessary for the following sections.

2.1 Complex projective space

The notion of projective space is vital to Schubert calculus, intuitively, projective space is the set of lines in a vector space, a more rigorous definition is:

Definition 2.1. *For a field $\mathbb{F}$ the projective space $\mathbb{P}(\mathbb{F}^n)$ is the set of equivalence classes of $\mathbb{F}^{n+1} \setminus \{0\}$ under the equivalence relation $x \sim y \iff x = \lambda y, \lambda \in \mathbb{F}^\times$.*

Elements of $\mathbb{P}^n(\mathbb{F})$ can be written as classes $[\lambda_1, \lambda_2, \dots, \lambda_{n+1}]$ such that not all coefficients λ_i are 0 and where two classes $[\lambda_1, \dots, \lambda_{n+1}] = [\lambda\lambda_1, \dots, \lambda\lambda_{n+1}]$ with $\lambda \in \mathbb{F}^\times$ are the same.

The notion of projective space is well defined for any field, but here the focus will be on the case of complex projective space $\mathbb{P}^n(\mathbb{C})$, for convenience we will denote this $\mathbb{P}^n$.

Projective space can also be viewed as an analogue of affine space with an added point at infinity, where parallel lines intersect at infinity. This perspective will be useful when considering the number of intersections of different objects later on.

Algebraic geometry posits that the closed sets of the Zariski topology on affine space are in correspondence with radical ideals in a polynomial ring, of

which they are the zero-locus. A similar Zariski topology exists on projective space, only the ideals must not only be radical but also homogeneous, since scaling all entries by a non-zero scalar does not change the point in question and so should not change whether the polynomial has a 0 at that point.

2.2 The Grassmannian and the Plücker Embedding

We defined complex projective space as equivalence classes of 1-dimensional subspaces of some given complex vector-space. A possible generalization of this is the set of k -dimensional subspaces of a vector space $\mathbb{C}^n$.

Definition 2.2. *The Grassmannian $\text{Gr}(k, \mathbb{C}^n)$ or simply $\text{Gr}(k, n)$ is the set of k -dimensional subspaces of $\mathbb{C}^n$.*

When $k = 1$, the identity map sends $\text{Gr}(1, n)$ to $\mathbb{P}^{n-1}$. There is a way to generalize this embedding to higher dimensional Grassmannian.

Definition 2.3. *A Plucker embedding is a map embedding a Grassmannian into projective space:*

$$\text{Gr}(k, n) \rightarrow \mathbb{P}^{\binom{n}{k}-1}, \langle v_1, v_2, \dots, v_k \rangle \mapsto v_1 \wedge v_2 \wedge \dots \wedge v_k$$

From a more concrete perspective, a k -dimensional space in $\mathbb{C}^n$ can be seen as the space generated by the columns of an $n \times k$ matrix. Such a matrix has $\binom{n}{k}$ $k \times k$ minors. Taking the determinants of these $k \times k$ minors gives a vector in $\mathbb{C}^{\binom{n}{k}}$. Since choosing a basis (i.e. multiplying by an invertible matrix) scales all the determinants by a nonzero scalar, two matrices generating the same space will have the same image in $\mathbb{P}^{\binom{n}{k}-1}$, the map is therefore well defined.

The objective of this embedding into projective space is to establish some geometric structure on the Grassmannian.

Proposition 2.1. *The Plucker embedding endows the Grassmannian with the structure of an irreducible projective variety.*

Proof. A single point in $\text{Gr}(k, n)$ is an irreducible projective variety. The group $GL_n(\mathbb{C})$, with its usual action on vector spaces, acts transitively on $\text{Gr}(k, n)$, and is itself irreducible. The Grassmannian is therefore an irreducible variety, being the $GL_n(\mathbb{C})$ -orbit of an irreducible variety. $\square$

Example. Consider the 2 dimensional subspaces of 4 dimensional complex space, $\text{Gr}(2, 4)$. The elements of these can be viewed as the column-spans of matrices

$$\begin{pmatrix} a & b \\ c & d \\ e & f \\ g & h \end{pmatrix}.$$

The Plucker embedding sends this to $[ad - bc, af - be, ah - bg, cf - de, ch - dg, eh - fg] \in \mathbb{P}^5$, which is in the zero locus of the homogeneous irreducible

polynomial $X_1X_6 - X_2X_4 + X_3X_4$. Conversely, any element in the zero locus of this polynomial can be written as the image of some 2-dimensional space, this is easy to see by scaling one of the entries to 1 (without loss of generality since at least one entry is non-zero), then the point is of the form $[1, a, b, c, d, ad - bc]$, which is the image of a row-reduced matrix. This gives that the image of $\text{Gr}(2, 4)$ is an irreducible variety.

Note. *Examples using $\text{Gr}(2, 4)$ will appear throughout this report, because this case corresponds to the set of lines in 3-dimensional projective space, which is our motivating problem.*

3 The flag variety

In $\mathbb{C}^n$, consider sequences of subspaces $(V_i)_{i=1}^n$, such that $\dim(V_i) = i$ and $V_i \subset V_{i+1} \forall i \in \{1, 2, \dots, n-1\}$. We call such sequences complete flags.

The group $GL_n(\mathbb{C})$ acts transitively on the set of complete flags. To see this we can choose vectors $v_i \in V_i \setminus V_{i-1}$, then each V_i is the space generated by $\langle v_1, \dots, v_i \rangle$, since GL_n can send these vectors to any other linearly independent family, any sequence of subspaces can arise, and so the GL_n -orbit of any flag is simply the set of all flags. The stabilizers in GL_n of some flag $\langle v_1 \rangle \subset \langle v_1, v_2 \rangle \subset \dots$ are the linear maps which send each family of vectors into the subspace generated by that family, this is exactly the upper triangular matrices written in the basis $(v_1, v_2, \dots, v_n)$, which send each vector v_i in the basis to a vector in the space $\langle v_1, \dots, v_i \rangle$.

We can therefore identify the set of flags with the quotient G/B of invertible linear maps $G = GL_n(\mathbb{C})$ modulo the set of upper triangular matrices B . This also gives a natural way of reading the subspaces V_i in a flag as the spaces generated by the first i columns of some coset of B in G .

3.1 The Plücker embedding on flags

Complete flags are sequences of subspaces of increasing dimension, so there is a natural embedding $V_1 \subset V_2 \subset \dots \subset V_{n-1} \mapsto (V_1, V_2, \dots, V_{n-1})$ of the set of flags into the Cartesian product of Grassmannians $\prod_{i=1}^{n-1} \text{Gr}(i, \mathbb{C}^n)$. The Plücker embedding then provides an embedding of the set of flags into projective space via the map:

$$\begin{array}{ccccccc} G/B & \hookrightarrow & \text{Gr}(1, \mathbb{C}^n) & \times \dots \times & \text{Gr}(i, \mathbb{C}^n) & \times \dots \times & \text{Gr}(n-1, \mathbb{C}^n) \\ & & \downarrow & & \downarrow & & \downarrow \\ & & \mathbb{P}^{n-1}(\mathbb{C}) & \times \dots \times & \mathbb{P}^{\binom{n}{i}-1}(\mathbb{C}) & \times \dots \times & \mathbb{P}^{n-1}(\mathbb{C}). \end{array}$$

The image of this map is contained in the image of the Grassmannian on each projective space, but it is the subset of the product of these Grassmannians that has the property that each subspace is contained in the next, this property

$w \in S_3$	$\theta_w \in G \backslash (G/B \times G/B)$		
	Lines	Planes	Intersection
()	$V_1 = U_1,$	$V_2 = U_2$	
(12)	$V_1 \neq U_1,$	$V_2 = U_2$	
(23)	$V_1 = U_1,$	$V_2 \neq U_2,$	$V_2 \cap U_2 = V_1 = U_1$
(123)	$V_1 \neq U_1,$	$V_2 \neq U_2,$	$V_2 \cap U_2 = V_1$
(132)	$V_1 \neq U_1,$	$V_2 \neq U_2,$	$V_2 \cap U_2 = U_1$
(13)	$V_1 \neq U_1,$	$V_2 \neq U_2,$	$V_2 \cap U_2 \notin \{V_1, U_1\}$

Table 1: Configurations of $(V_1 \subset V_2)$ and $(U_1 \subset U_2)$ in 3-dimensional space with associated permutations in S_3 .

can be translated to the fact that the Plucker coordinates in adjacent projective spaces should satisfy some homogeneous polynomial equations. The fact that G/B is the zero-locus of this set of polynomials in turn gives that the image of the set of flags is a closed set, and therefore that it has the structure of an algebraic variety. It can also be shown that G/B is a irreducible projective variety by the fact it is the GL_n -orbit of a single point, as with the Grassmannian.

Definition 3.1. *The image of G/B by the Plucker embedding is called the flag variety.*

3.2 Flags in relative position

For the rest of this report we will fix some standard basis $(v_1, \dots, v_n)$ of $\mathbb{C}^n$ and associate the standard flag $V_1 \subset V_2 \subset \dots \subset V_{n-1} \subset \mathbb{C}^n$, where each $V_i = \langle v_1, \dots, v_i \rangle$ is the space generated by the first i vectors of the standard basis.

Since we are interested in questions about intersections we would like to study the orbits of two flags in a certain relative position to each other. That is, we are interested in the orbits of G acting on the set $G/B \times G/B$, where the action of G is the diagonal action given by $g \mapsto ((b_1, b_2) \mapsto (gb_1, gb_2))$. These orbits give the possible configurations of two flags in space. For instance, in three dimensional space, flags consist of a plane and a line living in the plane, the orbits of the double flag variety are described in table 1.

The action of G on the double flag, intuitively, can send each flag anywhere in space, only preserving some relationship between the flags. Since the defining characteristic of the orbits is the relationship between the flags and not the precise position of the flags themselves, fixing the first flag in some set position (say, the standard flag) and studying the position of the second flag is equivalent to studying the double flag configuration. The members of G which fix the standard flag are exactly the upper triangular matrices B . Studying the G -orbits of double flags is therefore tantamount to studying the B -orbits of flags. Formally, the following is a bijection between the G -orbits of $G/B \times G/B$ and

the B -orbits of G/B :

$$\begin{aligned} G \backslash (G/B \times G/B) &\rightarrow B \backslash (G/B) \\ G(g_0B, g_1B) &\mapsto B(g_0^{-1}g_1B) \\ G(B, g_1B) &\hookleftarrow B(g_1B) \end{aligned}$$

Definition 3.2. *The images of the B -orbits of G/B are called Schubert cells.*

The Schubert cells are given by equivalence classes of matrices of the form BgB with $g \in G$. Consider the standard basis $(v_1, \dots, v_n)$ of an n -dimensional space. The flag gB can be associated to an ordered basis $(u_1, \dots, u_n)$, where the i th subspace of the flag is the subspace generated by the first i basis vectors. Adding some linear combination of u_k with $k < i$ to u_i will not change the space generated by the first i vectors and so will represent the same flag. In particular, letting $p(i)$ be the greatest integer such that $v_{p(i)}$ has nonzero coefficient in u_i , then all subsequent vectors u_k , $k > i$ in the basis can be replaced with vectors with 0 coefficient $v_{p(i)}$ by B -action. Proceeding via finite induction in this way for each $p(i)$ gives a basis corresponding to the flag for which no two basis vectors have the same greatest non-zero coefficient component vector in the standard basis $(v_i)_{i=1}^n$. This can be assimilated to a restricted version of Gaussian elimination, where one can replace a column with linear combinations of itself and all columns to its left. Seen as the columns of a matrix, this means that the action of upper triangular matrices in B can always provide a matrix for which the rows have pairwise distinct first non-zero entry, that is to say, a permutation of an upper triangular matrix Bw where w is the matrix representation of some permutation in S_n . In other words, any matrix in G can be written as BwB for some permutation $w \in S_n$. Furthermore the permutation w is given by the construction of p in the proof above i.e. $w(i) = p(i)$, $\forall 1 \leq i \leq n$. Since this is invariant under the action of B , the permutation is unique for each B -orbit. This gives the following result:

Proposition 3.1. *The Schubert cells in n -dimensional space are indexed by the members of the group of permutations S_n . The flag variety admits a decomposition as a disjoint union of Schubert cells $G/B = \sqcup_{w \in S_n} BwB/B$.*

We will denote $\theta_w = BwB \cong \theta'_w = G(B, wB)$ the Schubert cell indexed by $w \in S_n$.

The Schubert cell θ_w is the B -orbit of the flag $\langle v_{w(1)} \rangle \subset \langle v_{w(1)}, v_{w(2)} \rangle \subset \dots \subset \langle v_{w(1)}, \dots, v_{w(i)} \rangle \subset \dots \subset \mathbb{C}^n$, where the i th subspace is generated by the first i vectors of the standard basis permuted by w .

3.3 The Bruhat order on the flag variety

This subsection is concerned with the inclusion of Schubert cells in the closures (with respect to the Zariski topology) of others and a partial order on the group of permutations induced by these inclusions. This will be useful later on when describing the topological structure of the flag variety and computing its cohomology groups.

3.3.1 The inclusion order on Schubert cells

We have shown that the flag variety G/B admits a decomposition as $\sqcup_{w \in S_n} \theta_w$, where $\theta_w = BwB$ are Schubert cells. The closure $\overline{\theta_w}$ of a Schubert cell θ_w contains some other cells θ_u , in fact, the closure is stable by the action of B , so it must be the union of some collection of Schubert cells. The following partial order characterizes these inclusions.

Definition 3.3. *The partial order $\leq_1$ is defined by $u \leq_1 w \iff \theta_u \subset \overline{\theta_w} \iff \theta'_u \subset \overline{\theta'_w}$ respectively.*

First, we must prove that this definition is consistent, which comes from the statement of the following lemma:

Lemma 3.1. *The inclusions of closures on $B \backslash G/B$ and $G \backslash (G/B \times G/B)$ are the same, i.e. $\theta'_u \subset \overline{\theta'_w} \iff \theta_u \subset \overline{\theta_w}$.*

Proof. The implication $\theta_u \subset \overline{\theta_w} \implies \theta'_u \subset \overline{\theta'_w}$ relies on the fact that the map taking G/B to $G \backslash (G/B \times G/B)$ is the inclusion map $V \mapsto (B, V)$. If a polynomial vanishes on the image of (B, W) for all $W \in \theta_w$, and $\theta_u \subset \overline{\theta_w}$, then it must also vanish on (B, U) for all $U \in \theta_u$, thus $\theta'_u \subset \overline{\theta'_w}$ by taking the G -orbit. $\square$

For $\theta'_u \subset \overline{\theta'_w} \implies \theta_u \subset \overline{\theta_w}$, consider the ideal I of polynomials vanishing on the closed image of (B, G) by the Plucker embedding. The union of I with the ideal of $G \backslash (g_0 B, g_1 B)$ vanishes only on the image of the subset of elements of the form $(B, g_0^{-1} g_1 B)$, since this depends only on the $g_0^{-1} g_1 B$, which is its image in $B \backslash G/B$, thus the inclusion of ideals is preserved by the mapping and the implication holds.

3.3.2 The Bruhat order on S_n

The order discussed above is an order on the group of permutations, but it is difficult to see what the order implies about the permutations, outside the context of the flag variety. To better understand this, it is helpful give another, group-theoretic definition of the above order.

Definition 3.4. *The Bruhat order $\leq_2$ on S_n is defined by $u \leq_2 w$ if there is an adjacent transposition $(i \ i+1)$ such that $w = u(i \ i+1)$ and $\ell(u) < \ell(w)$, that is, the minimum number of adjacent transpositions needed to write u is less than the minimum number for w .*

Definition 3.5. *For some permutation $w \in S_n$, we write $w[i, j] = |\{a \leq i | w(a) \geq j\}|$.*

Lemma 3.2. *$u \leq w$ if and only if $u[i, j] \leq w[i, j]$ for all $1 \leq i, j \leq n$.*

Proof. The proof is provided in theorem 2.1.5 of [2]. $\square$

Lemma 3.3. *For a permutation $u \in S_n$ and associated flag $U_1 = \langle v_{u(1)} \rangle \subset U_2 = \langle v_{u(1)}, v_{u(2)} \rangle \subset \dots$, consider the standard flag $V_1 = \langle v_1 \rangle \subset V_2 = \langle v_1, v_2 \rangle \subset \dots$ we have $u[i, j] = n - \dim(U_i \cap V_{j-1})$.*

Proof. Since both spaces U_i and V_{j-1} are expressed in terms of vectors in the standard basis, their intersection is the space generated by the basis vectors they have in common. This corresponds to the vectors v_k such that $v_k \in U_i, v_k \in V_{j-1} \iff k = u(k')$ where $k' \leq i$ and $u(k') < j$. looking at the definition of the number $u[i, j]$ we see that this is precisely the converse relation, and thus there are $n - u[i, j]$ such basis vectors u_k and that is the dimension of the space $\dim(U_i \cap V_{j-1})$. $\square$

Theorem 3.1. *The partial orders $\leq_1$ and $\leq_2$ on S_n are the same.*

Proof. First we must prove that if $w = u(i \ i+1)$ with $\ell(u) < \ell(w)$ then $\theta_u \subset \overline{\theta_w}$. This means in particular that $w(i) > u(i), w(i+1) = u(i), w(i) = u(i+1)$, and for all other $j \notin \{i, i+1\}, w(j) = u(j)$. Let $(v_1, \dots, v_n)$ be the standard basis and consider the flags in θ_u and θ_w , except from the i -th spaces, the spaces in the flags of these two classes are the same, i.e. θ_u and θ_w are the B -orbits of the flags $U_1 = \langle v_{u(1)} \rangle \subset \dots \subset U_i = \langle v_{u(1)} \dots v_{u(i)} \rangle \subset \dots \subset \mathbb{C}^n$ and $W_1 = \langle v_{w(1)} \rangle \subset \dots \subset W_i = \langle v_{w(1)} \dots v_{w(i)} \rangle \subset \dots \subset \mathbb{C}^n$ respectively, where $U_k = W_k, \forall k \neq i$. Since $w(i) > u(i)$ the B -orbit of the space $\langle v_{w(1)} \dots v_{w(i)} \rangle$ is the B -orbit of $\langle v_{w(1)} \dots v_{w(i-1)}, \alpha v_{w(i)} + v_{v(i)} \rangle$ for any $\alpha \in \mathbb{C} \setminus \{0\}$. Taking the limit as $\alpha \rightarrow 0$ gives that

$$(W_1 \subset W_2 \subset \dots \subset U_i \subset \dots) = (U_1 \subset U_2 \subset \dots \subset U_i \subset \dots) \in \overline{\theta_w}.$$

Taking the B -orbits gives that $\theta_u \subset \overline{\theta_w}$. $\square$

Now to prove the reverse implication, assume that $\theta_u \subset \overline{\theta_w}$. Consider the number $d = w[i, j]$, any matrix M generating W_i , and the column matrix

$$A = \begin{pmatrix} v_1 & | & v_2 & | & \dots & | & v_{j-1} & | & M \end{pmatrix}.$$

Since the dimension of the intersection $V_{j-1} \cap W_i$ is $n - d$ by lemma 3.3, the dimension of the space generated by the columns of the matrix is $d' = i + j - 1 - n + d$ then all determinants of minors of the matrix with size $d' + 1$ will vanish. This gives a polynomial condition on the i -th space W_i , since $\theta_u \subset \overline{\theta_w}$, U_i must also satisfy this condition and so the dimension of the intersection $V_{j-1} \cap W_i$ must be at least d . this means that $u[i, j] \leq w[i, j]$. Since this is true for all i, j , lemma 3.2 provides that $u \leq w$.

Corollary 3.1. *The closure of a Schubert cell θ_w can be given as a disjoint union $\overline{\theta_w} = \sqcup_{u \leq w} \theta_u$.*

Proof. The cell closure $\overline{\theta_w}$ is stable by B -orbit, so it must be given as a disjoint union of B -orbits. The above proposition tells us the members of this union are the cells θ_u such that $u \leq w$. $\square$

We call the closure $\overline{\theta_w}$ of a Schubert cell θ_w a Schubert variety. Schubert varieties are closed subsets of the projective variety G/B , so they each have the structure of a projective variety.

4 The Cohomology of Schubert Cells

This report will not deal with the fundamental results of homology and cohomology, which are laid out in [5]. The results that are important to Schubert calculus are those concerning cellular homology and the cup product. Cellular homology gives an easy description of the homology groups of CW-complexes, of which the flag variety is one. The cup product is the product for the cohomology ring, and allows for the computation of intersection problems explicitly. The a th homology group of the flag variety will be denoted $H_a(G/B, \mathbb{Z})$ or simply $H_a(G/B)$, the a th cohomology group denoted $H^a(G/B)$, and the cohomology ring denoted $H^*(G/B)$.

4.1 The CW-structure of Schubert Cells

The objective of this subsection is to prove that the decomposition of the flag variety into Schubert cells endows it with the structure of a CW-complex. In other words, to prove that each Schubert cell θ_w is homeomorphic to an open ball of dimension a and that its boundary $\partial\theta_w = \overline{\theta_w} - \theta_w = \sqcup_{u < w} \theta_u$ glues to the union of all cells with smaller dimension than a , denoted X_{a-1} via a continuous map $\sqcup_{u < w} \theta_u \rightarrow X_{a-1}$. To show that the Schubert cells are homeomorphic to open balls, it is convenient to consider them as spaces spanned by families of matrices with a certain amount of free entries. In particular the row echelon form $\theta_w = BwB$

$$BwB = \begin{pmatrix} 0 & \dots & 0 & 0 & 1 & * & \dots & * \\ 0 & \dots & 1 & * & 0 & * & \dots & * \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & \dots & 0 & * & 0 & * & \dots & * \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & 1 & 0 & 0 \end{pmatrix} B$$

where $*$ represents a free entry.

Given the number of free entries a , the family of matrices is homeomorphic to affine space of dimension a , which itself is homeomorphic to an open ball of dimension a . The number of free entries, and therefore the dimension of the open ball, is determined by the permutation w in the following way.

Lemma 4.1. *The cell θ_w is homeomorphic to an open ball of dimension equal to the cardinality of the set $S_w = \{(i, j) \in [1, n] \mid i < j, w(i) > w(j)\}$.*

Proof. The proof can be done as above by induction with respect to adjacent permutations. When $u = ()$ is the identity permutation, the associated matrix

is the identity matrix and so there are no free entries, so θ_u is homeomorphic to a 0-ball, i.e. a point. Suppose the lemma is true for some $u \in S_n$ and $w = (i \ i+1)u$ where $w(i) > w(i+1)$. Then the cardinality of the set S_w from the statement of the lemma is 1 more than S_u , since no relations are changed between u and w except for the relation $u(i) < w(i+1)$, $w(i) > w(i+1)$. The matrix Bw has the same shape as Bu except that the columns i th and $i+1$ th rows are interchanged, since $u(i) < u(i+1)$ this means that in Bu the i th row creates a 0 at the $u(i)$ th entry of the $i+1$ th row. The matrix for w is exactly the same except that the rows are switched so that this zero is not created and therefore there is one extra free entry. Proceeding by induction proves the statement of the lemma, since all permutations can be expressed as a product of adjacent transpositions. $\square$

The proof by adjacent permutations means in particular that if $u < w$ in the Bruhat order, then the dimension of θ_u is less than θ_w . This means that all the cells in the boundary $\partial\theta_w = \sqcup_{u < w} \theta_u$ have lower dimension than $a = \dim(\theta_w)$, and so are included in the $a-1$ skeleton of G/B , X_{a-1} . Therefore, the inclusion map $\iota : \sqcup_{u < w} \theta_u \rightarrow X_{a-1}$ provides a continuous gluing map for the boundary of each cell.

Therefore the flag variety has the structure of a CW-complex, and all the cells have even real dimension, since they are homeomorphic to affine complex space. This means that the homology group of the flag variety has a basis given by the Schubert cells, since all the boundary maps are the 0 map (since all the odd homology groups $H_{2a+1}(G/B)$ are empty).

Theorem 4.1. *The cohomology group $H^*(G/B)$ is a free group with basis given by any Schubert cell decomposition of G/B . The cohomology groups $H^{2a}(G/B)$ are free groups with basis given by the Schubert cells with complex co-dimension a .*

Proof. The structure of the cohomology groups is the same homology groups, since these are all free abelian. $\square$

We call the homology class $[\theta_w]$ of a Schubert cell θ_w a Schubert class, this class is independent of the chosen basis/flag, and only depends on the Schubert cell, i.e. the permutation w .

4.2 The cup product on Schubert Cells

The above cellular structure of the flag variety provides the cellular homology of the flag variety, each homology group H_a is a free abelian group with the a -cells of the flag variety as a basis. The cohomology groups H^a have the same group structure as the homology groups, but have an additional ring structure provided by the cup product $\smile : H^i \times H^j \rightarrow H^{i+j}$. The following theorem characterizes the cup product on Schubert classes.

Theorem 4.2. *Let $[X], [Y]$ be Schubert classes. There is a open dense subset $\Omega \subset G$ such that $\forall g \in \Omega$, $[X][Y] = [X \cap gY]$.*

Proof. This is a special case of theorem 2.21 in [8]. $\square$

Notably, since Ω is dense in G the expression $gY, \forall g \in \Omega$ represents a Schubert cell in 'general position'. This cup product allows solving intersection problems, since the product itself gives the cohomology class of the intersection between the two objects in general position.

Schubert calculus consists in translating the geometric conditions defining objects into the language of cohomology, and then using the cup product on the cohomology classes to compute the intersections of the geometric objects and their multiplicities.

5 Schubert Calculus of Grassmannians

To study the question of the number of lines intersecting four lines in general position, it is useful to specialize to the study of Grassmannians, and the specific example seen above of $\text{Gr}(2, 4)$.

5.1 Schubert Cells of Grassmannians

There is a natural projection from the flag variety to a Grassmannian $\text{Gr}(k, n)$ by sending a complete flag $V_1 \subset V_2 \subset \dots \subset V_k \subset \dots \subset \mathbb{C}^n$ to its k th entry V_k . This map is clearly surjective but not injective. The Grassmannian can also be decomposed into its B -orbits, which are also called Schubert cells, but since the map from the flag variety to the Grassmannian is not injective, several Schubert cells of the flag may be mapped to the same cell in the Grassmannian. In particular, consider a permutation w and the associated flag $\langle v_{w(1)} \rangle \subset \dots \subset \langle v_{w(1)}, \dots, v_{w(k)} \rangle \subset \dots$. Since the image of this flag is simply $\langle v_{w(1)}, \dots, v_{w(k)} \rangle$, composing whh' where h is any permutation of the first k values and h' is any permutation of the last $n - k$ values will not change the resulting space in the Grassmannian. Let $S([1, k])$ be the group of permutations on the integers between 1 and k and let $S([k + 1, n])$ be the group of permutations on the integers between $k + 1$ and n and let $S = S_n / S([1, k])S([k + 1, n])$ be the quotient group, then the following proposition gives the Schubert cell decomposition of the Grassmannian.

Proposition 5.1. *The Grassmannian is the disjoint union of Schubert cells $\text{Gr}(k, n) = \sqcup_{s \in S} \theta_s$, where θ_s is the B -orbit of $\langle v_{w(1)}, \dots, v_{w(k)} \rangle$, where w is any representative of s .*

The Schubert cells of the Grassmannian can also be seen as the P -orbits of

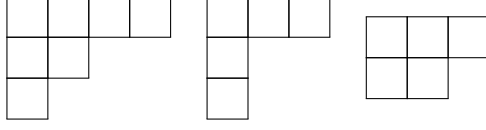


Figure 1: Young diagrams $(4 \geq 2 \geq 1)$, $(3 \geq 1 \geq 1)$, and $(3 \geq 2)$.

G/B , where

$$P = \left\{ \begin{pmatrix} * & \dots & * & * & \dots & * \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ * & \dots & * & * & \dots & * \\ 0 & \dots & a_{k+1,k} = 0 & * & \dots & * \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & a = 0 & * & \dots & * \end{pmatrix} \in GL_n(\mathbb{C}) \mid * \text{ is a free entry} \right\}.$$

This comes directly from the product of the subgroups of the symmetric group with the group of upper triangular matrices B .

In particular the above proposition gives the number of Schubert cells of the Grassmannian:

Corollary 5.1. *There are $\frac{n!}{k!(n-k)!} = \binom{n}{k}$ Schubert cells in the decomposition of the Grassmannian $\text{Gr}(k, n)$.*

Proof. This is the order of $S = S_n / (S([1, k])S([k+1, n]))$ since $(S([1, k])S([k+1, n])) \cong S([1, k]) \times S([k+1, n])$ has order $k!(n-k)!$. $\square$

This should be intuitive, since each Schubert cell is the B -orbit of some space $\langle v_{w(1)}, \dots, v_{w(k)} \rangle$ generated by k vectors of the standard basis, and there are $\binom{n}{k}$ different ways of choosing k basis vectors from the standard basis.

5.2 Young diagrams

A nice way of classifying Schubert cells of the Grassmannian is with Young diagrams. Young diagrams are finite collections of boxes arranged in columns with non increasing number of boxes per column. In other words a Young diagram λ is a sequence of non-negative integers $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_i$. We say that a Young diagram “fits inside a $j \times k$ box” if $j \geq \lambda_1$ and $i \leq k$, i.e. there are at most k rows and each row has at most j boxes. We will denote the total number of boxes of a Young diagram by $|\lambda|$.

The complement of a Young diagram $\lambda = (\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_i)$ in an $j \times k$ box is the Young diagram $\hat{\lambda}$ obtained by taking the empty boxes and rotating these 180 degrees, in other words $\hat{\lambda} = (j - \lambda_i \geq j - \lambda_{i-1} \geq \dots \geq j - \lambda_2 \geq j - \lambda_1)$.

Lemma 5.1. *There is a bijection between the Schubert cells of $\text{Gr}(k, n)$ and the set of Young diagrams fitting into a $(k) \times (n - k)$ box.*

Proof. All that really needs to be proved here is that there are $\binom{n}{k}$ Young diagrams fitting into a $(k) \times (n - k)$ box. The symbol $\binom{n}{k}$ can be viewed as the number of distinct sets of k numbers between 1 and n , this can in turn be written as the number of distinct sequences $(a_i)_{i=1}^n$ such that $1 \leq a_1 < a_2 < \dots < a_k \leq n$. Young diagrams fitting into a $(k) \times (n - k)$ box can be viewed as sequences of numbers $(\lambda_i)_{i=1}^k$ such that $n - k \geq \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k \geq 0$. Then the following is a bijection between these two constructions:

$$\begin{aligned} 1 \leq a_1 < \dots < a_k \leq n &\rightarrow n - k \geq \lambda_1 \geq \dots \geq \lambda_k \geq 0 \\ a_i &\mapsto n - k + i - a_i \\ n - k + i - \lambda_i &\leftarrow \lambda_i. \end{aligned}$$

□

Composing with the map that sends the Schubert cell $B\langle v_{w(1)}, \dots, v_{w(k)} \rangle$ to $\{w(i) | 1 \leq i \leq k\}$ associates each Schubert cell to a specific Young diagram. In fact the Young diagram associated in this way gives the shape of the 0s of the reduced form of the matrix with rows $v_{w(i)}$ for $1 \leq i \leq k$ (excluding the $k - 1$ 0s in each column). Another description is that the Young diagram is the complement of the diagram given by the shape of the free entries in such a matrix. These two perspectives are compared for $\text{Gr}(2, 4)$ in table 2. The fact that the associated Young diagram is the complement of the shape of the free entries also characterizes the relationship between the dimension of the Schubert cell and the number of boxes in its associated Young diagram.

Corollary 5.2. *Let θ_s be a Schubert cell of the Grassmannian $\text{Gr}(k, n)$ with dimension d , then it is associated with a Young diagram containing $k(n - k) - d$ boxes.*

We denote X_λ the Schubert cell that is mapped to the Young diagram λ . This correspondence between the Young diagrams and Schubert cells will be useful for computing the product of Schubert classes. The computation rules will be laid out in the following subsection.

5.3 The Pieri Rule and Poincaré Duality

In this subsection it is assumed that, working in $\text{Gr}(k, n)$, all Young diagrams fit into a $k \times (n - k)$ box, otherwise we associate the Young diagram to 0 in the cohomology ring.

The product $\sigma = [X_\lambda][X_\mu]$ is a sum of elements of codimension $|\lambda| + |\mu|$ by the definition of the cup product, computing the structure of the cohomology ring just involves finding the coefficients of $[X_\delta]$ in σ where δ is a Young tableau of size $|\delta| = |\lambda| + |\mu|$.

Proposition 5.2. *(Poincaré Duality for Schubert cells) Let λ and μ be two Young diagrams in a $k \times (n - k)$ box such that $|\lambda| + |\mu| = k(n - k)$. Let $[X_\lambda]$ and $[X_\mu]$ be the associated Schubert classes in $\text{Gr}(k, n)$, then $[X_\lambda][X_\mu]$ is 1 if μ is the complement of λ and 0 otherwise.*

Space	Reduced matrix	Mapping	Shape of *s	Shape of 0s
$\langle v_1, v_2 \rangle$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$	$(1 < 2) \mapsto (2 \geq 2)$	$\emptyset$	
$\langle v_1, v_3 \rangle$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & * & 1 & 0 \end{pmatrix}$	$(1 < 3) \mapsto (2 \geq 1)$		
$\langle v_1, v_4 \rangle$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & * & * & 1 \end{pmatrix}$	$(1 < 4) \mapsto (2 \geq 0)$		
$\langle v_2, v_3 \rangle$	$\begin{pmatrix} * & 1 & 0 & 0 \\ * & 0 & 1 & 0 \end{pmatrix}$	$(2 < 3) \mapsto (1 \geq 1)$		
$\langle v_2, v_4 \rangle$	$\begin{pmatrix} * & 1 & 0 & 0 \\ * & 0 & * & 1 \end{pmatrix}$	$(2 < 4) \mapsto (1 \geq 0)$		
$\langle v_3, v_4 \rangle$	$\begin{pmatrix} * & * & 1 & 0 \\ * & * & 0 & 1 \end{pmatrix}$	$(3 < 4) \mapsto (0 \geq 0)$		$\emptyset$

Table 2: The correspondence between Schubert cells of $\text{Gr}(2, 4)$ and Young diagrams in a 2×2 box.

The only possible class in the product is the class of maximal codimension, so the class of a point, Poincaré duality gives the multiplicity of the points of intersection, which turns out to be either 1 or 0.

The complete proof of this proposition is given in proposition 2.22 of [8].

Theorem 5.1. (*Pieri Rule for Schubert cells*) Let (m) be a Young diagram consisting of one line with m boxes and λ be another Young diagram. Then $[X_{(m)}][X_\lambda] = \sum_{\mu \in m \otimes \lambda} X_\mu$, where $m \otimes \lambda$ is the set of all Young diagrams that fit into a $k(n-k)$ box that can be made by adding m boxes to λ , with no two boxes in the same column.

Proof. We know that the product $[X_{(m)}][X_\lambda] = \sum_{|\mu|=|\lambda|+m} c_\mu \mu$ where c_μ is a coefficient in $\mathbb{Z}$. It is possible to find the coefficient c_μ of μ in this sum using Poincaré duality, by multiplying $[X_{(m)}][X_\lambda]$ by the Schubert class corresponding to the complement Young diagram $\hat{\mu}$ of μ . Then $[X_{(m)}][X_\lambda][X_{\hat{\mu}}] = c_\mu$. The proof of theorem 2.27 in [8] tells us that this product is 1 (i.e. the class of a point) when $n-k-\lambda_k \geq \hat{\mu}_1 \geq n-k-\lambda_{k-1} \geq \hat{\mu}_2 \geq \dots \geq n-k-\lambda_1 \geq \hat{\mu}_k$, and 0 otherwise. The above condition is the same as $\lambda_k \leq \mu_k \leq \lambda_{k-1} \leq \mu_{k-1} \leq \dots \leq \lambda_1 \leq \hat{\mu}_1$. This means two boxes cannot be added to the same column, since looking at the top row μ_t of this column, it will be greater than bottom the row λ_b with $t > b$, which contradicts the statement, conversely, if no two boxes are added to the same column, then the inequality holds. $\square$

Note. If μ does not fit into a $k(n-k)$ box, the $[X_\mu]$ is 0 in $H^*(\text{Gr}(k, n))$.

Young diagram	Cell	Notation	Geometric Meaning
	$B\langle v_1, v_2 \rangle$	$[X_0]$	Class of a line
	$B\langle v_1, v_3 \rangle$	$[X_1]$	Class of lines passing through a given point and contained in a given plane.
	$B\langle v_1, v_4 \rangle$	$[X_2']$	Class of lines passing through a point
	$B\langle v_2, v_3 \rangle$	$[X_2]$	Class of lines in a given plane
	$B\langle v_2, v_4 \rangle$	$[X_3]$	Class of lines intersecting some fixed line
$\emptyset$	$B\langle v_3, v_4 \rangle$	$[X_4]$	Class of all lines in space

Table 3: Schubert classes of $\text{Gr}(2, 4)$.

Examples. The Young diagrams in $1 \otimes (1)$ in a 2×2 box are and .

The Young diagrams in $2 \otimes (2 \geq 1)$ in a 3×3 box are , ,

and .

5.4 The 4 Lines Problem

To answer the question of how many lines intersect four lines in general position, it will first be useful to compute the cohomology ring of $\text{Gr}(2, 4)$, as well as describing the geometric situations in projective space which each Schubert class corresponds to. The geometric interpretations given in table 3 are derived from the intersections with the standard flag $\langle v_1 \rangle \subset \langle v_1, v_2 \rangle \subset \langle v_1, v_2, v_3 \rangle \subset \mathbb{C}^4$, which represent some fixed point, fixed line, and fixed plane in projective space. The multiplication table is derived using the rules in the above subsection and some basic algebra, it can be shown that the Pieri rule is sufficient to determine the multiplication table for every Grassmannian, with a correspondence to some ring of polynomials.

Armed with this multiplication table, the questions of lines intersecting 4 lines in general position is tantamount to calculating $[X_4]^4 = [X_4]^2([X_2] + [X_2']) = [X_4]([X_1] + [X_1]) = 2[X_0]$. This means there are exactly two lines intersecting 4 general lines in projective space.

Product	Justification
$[X_4]$ is the multiplicative identity	Pieri Rule
$[X_3]^2 = [X_2] + [X'_2]$	Pieri Rule
$[X_3][X_2] = [X_1]$	Pieri Rule
$[X_3][X'_2] = [X_1]$	Pieri Rule
$[X_3][X_1] = [X_0]$	Pieri Rule
$[X_3][X_0] = 0$	Pieri Rule
$[X_2]^2 = [X_0]$	Poincaré Duality
$[X_2][X'_2] = 0$	Poincaré Duality
$[X_2][X_1] = [X_3][X_2][X'_2] = 0$	$[X_1] = [X_3][X'_2]$
$[X'_2]^2 = [X_0]$	Poincaré Duality
$[X'_2][X_1] = [X_3][X_2][X'_2] = 0$	$[X_1] = [X_3][X_2]$
$[X_1]^2 = [X_3]^2[X_2]^2 = [X_3][X_3][X_0] = 0$	$[X_1] = [X_3][X_2]$
$[X_0][X_b] = 0 \quad \forall b \neq 4$	$[X_3][X_0] = 0$

Table 4: Multiplication table of the Schubert classes of $\text{Gr}(2, 4)$.

6 Schur Polynomials

In order to establish the fact that the Pieri rule determines the structure of the cohomology ring of the Grassmannian, it is necessary to find a correspondence with the ring of symmetric polynomials. The symmetric polynomials are the polynomials in $\mathbb{Z}[x_1, \dots, x_n]$ which are stable by the action of S_n , which, for each permutation w , sends x_i to $x_{w(i)}$. The set of symmetric polynomials is denoted $\mathbb{Z}[x_1, \dots, x_n]^{S_n}$.

6.1 Young Tableaux and Schur Polynomials

A Young tableau is a Young diagram filled with symbols, for our purposes they will always be filled with integers. If a Young tableau T can be obtained by filling the diagram λ , we say it is of shape λ .

Definition 6.1. *A Young tableau is called semi-standard if the entries are weakly increasing on each row and strictly increasing on each column.*

Consider a Young diagram λ , and all the semi-standard Young tableaux arising from filling λ with numbers between 1 and n . Let T be such a semi-standard Young diagram, and let t_i be the number of times the integer $1 \leq i \leq n$ appears in T , then $x^T = \prod_{i=1}^n x_i^{t_i}$ is the polynomial associated with T .

Definition 6.2. *The Schur polynomial associated to the Young diagram λ is the sum $s_\lambda(x_1, \dots, x_n) = \sum x^T$ where T runs over all possible semi-standard fillings of λ .*

These polynomials have a multiplication rule analogous to the Pieri rule for Young diagrams used in the multiplication of Schubert classes.

Theorem 6.1. (*Pieri Rule for Schur polynomials*) Let (m) be a Young diagram consisting of one row with m boxes and let λ be some other Young diagram. Then $s_\lambda s_{(m)} = \sum_{\mu \in \lambda \otimes m} s_\mu$.

Proof. It is sufficient to show that there is a correspondence between Young tableaux of shape $\lambda \otimes m$ and the cartesian product of $\lambda \otimes m$ that preserves the multiplicity of each integer entry. An algorithm for inserting a row into a semi-standard Young tableau such that no two added boxes are in the same column called “Jeu-de-Taquin” is given in [1] and corollary 2.3.19 of [3], the Jeu-de-taquin algorithm also provides a way to reverse this by turning the filling of the “extra” boxes into a row. Thus there is a bijection between the tableaux of shape $\lambda \otimes m$ and the pairs of tableaux of shape $(\lambda, (m))$ preserving multiplicity of entries. $\square$

Proposition 6.1. *The Schur polynomials $s_{(m)}$ of one-row Young diagrams (m) generate the ring of symmetric polynomials.*

Proof. The fundamental theorem of symmetric polynomials [8] gives that the collection of polynomials $h_m = \sum_{1 \leq i_1 \leq \dots \leq i_m \leq n} x_{i_1} \dots x_{i_m}$ for all $1 \leq m \leq n$ generate the ring $\mathbb{Z}[x_1, \dots, x_n]$. Then each $x_{i_1} \dots x_{i_m}$ corresponds exactly to a filling of the Young diagram (m) with the integers $(i_1 \leq \dots \leq i_m)$, thus the polynomials $s_{(m)}$ corresponding to the Young diagrams (m) generate the ring of symmetric polynomials. This in particular gives that the Pieri rule determines all multiplication in the ring. $\square$

Now it is possible to write down the surjective ring homomorphism

$$\phi : \mathbb{Z}[x_1, \dots, x_n]^{S_n} \rightarrow H^*(\text{Gr}(k, n)), \quad s_\lambda \mapsto \begin{cases} [x_\lambda] & \text{if } \lambda \text{ fits in a } k(n-k) \text{ box} \\ 0 & \text{otherwise.} \end{cases}$$

This is well defined because the Pieri rule determines all multiplication in the ring $\mathbb{Z}[x_1, \dots, x_n]^{S_n}$ and this Pieri rule is consistent with the Pieri rule on $H^*(\text{Gr}(k, n))$. The morphism is surjective since the classes $[X_\lambda]$ where λ is a Young diagram in a $k(n-k)$ box form a basis of the cohomology ring and these are the image of the polynomials s_λ . The surjectivity of this map means that the multiplication in the cohomology ring is also completely determined by the Pieri rule.

Note. This morphism also gives the multiplicities of multiplication in as an expression of the intersections between geometric objects, and therefore provides that products of Schur polynomials are given by sums of Schur polynomials with non-negative coefficients, since there cannot be negative amount of intersection.

Further work and Open Problems

We focused on computing the cohomology ring of the Grassmannian $H^*(\text{Gr}(kn))$ using the correspondence with Young tableaux and Schur polynomials. Computing the cohomology ring of the the flag variety $H^*(G/B)$ also admits a

combinatorial analogue, called pipe dreams, and a generalization of Schur polynomials called Schubert polynomials. Pipe dreams they give a combinatorial representation of the group S_n , and are described in [4], among others.

While the combinatorial analogue of the cohomology ring is understood for G/B and the Grassmannian, it is still not well understood for other matrix groups such as the orthogonal group or the special linear groups, some relatively recent work on the orthogonal case includes [9] and [10].

References

- [1] Per Alexandersson. *Operations on Young tableaux*. 2021. URL: <https://www.symmetricfunctions.com/tableauOperators.htm>.
- [2] Anders Björner and Francesco Brenti. *Combinatorics of Coxeter groups*. Graduate texts in mathematics. New York: Springer, 2005. ISBN: 9783540442387. URL: <https://sites.math.washington.edu/~billey/classes/reflection.groups/references/EntireBook.pdf>.
- [3] Lynne M. Butler. *Subgroup lattices and symmetric functions*. eng. Online-Ausg. American Mathematical Society. Providence, RI: American Mathematical Society, 1994. ISBN: 9781470401184.
- [4] Marcott Cameron. “On the Relationship between Pipe Dreams and Permutation Words”. In: (Sept. 2013). URL: <https://www.combinatorics.org/ojs/index.php/eljc/article/download/v20i3p40/pdf/>.
- [5] Allen Hatcher. *Algebraic topology*. Cambridge; New York: Cambridge University Press, 2002. ISBN: 9780521791601. URL: <https://pi.math.cornell.edu/~hatcher/AT/AT.pdf>.
- [6] Brion Michel. *Lectures on the geometry of flag varieties*. Oct. 2004.
- [7] Mustața Mircea. *Introduction to Algebraic Geometry*.
- [8] Evgeny Smirnov. “Grassmannians, flag varieties, and Gelfand-Zetlin polytopes”. In: arXiv:1508.03010 (Aug. 2015). arXiv:1508.03010. DOI: 10.48550/arXiv.1508.03010. URL: <http://arxiv.org/abs/1508.03010>.
- [9] Harry Tamvakis. “Schubert polynomials and Arakelov theory of orthogonal flag varieties”. In: arXiv:0907.3308 (Sept. 2013). DOI: 10.48550/arXiv.0907.3308. URL: <http://arxiv.org/abs/0907.3308>.
- [10] Harry Tamvakis. “Theta and eta polynomials in geometry, Lie theory, and combinatorics”. In: *First Congress of Greek Mathematicians* (2020), pp. 243–284. URL: <https://math.umd.edu/~harryt/papers/tepoglc.pdf>.